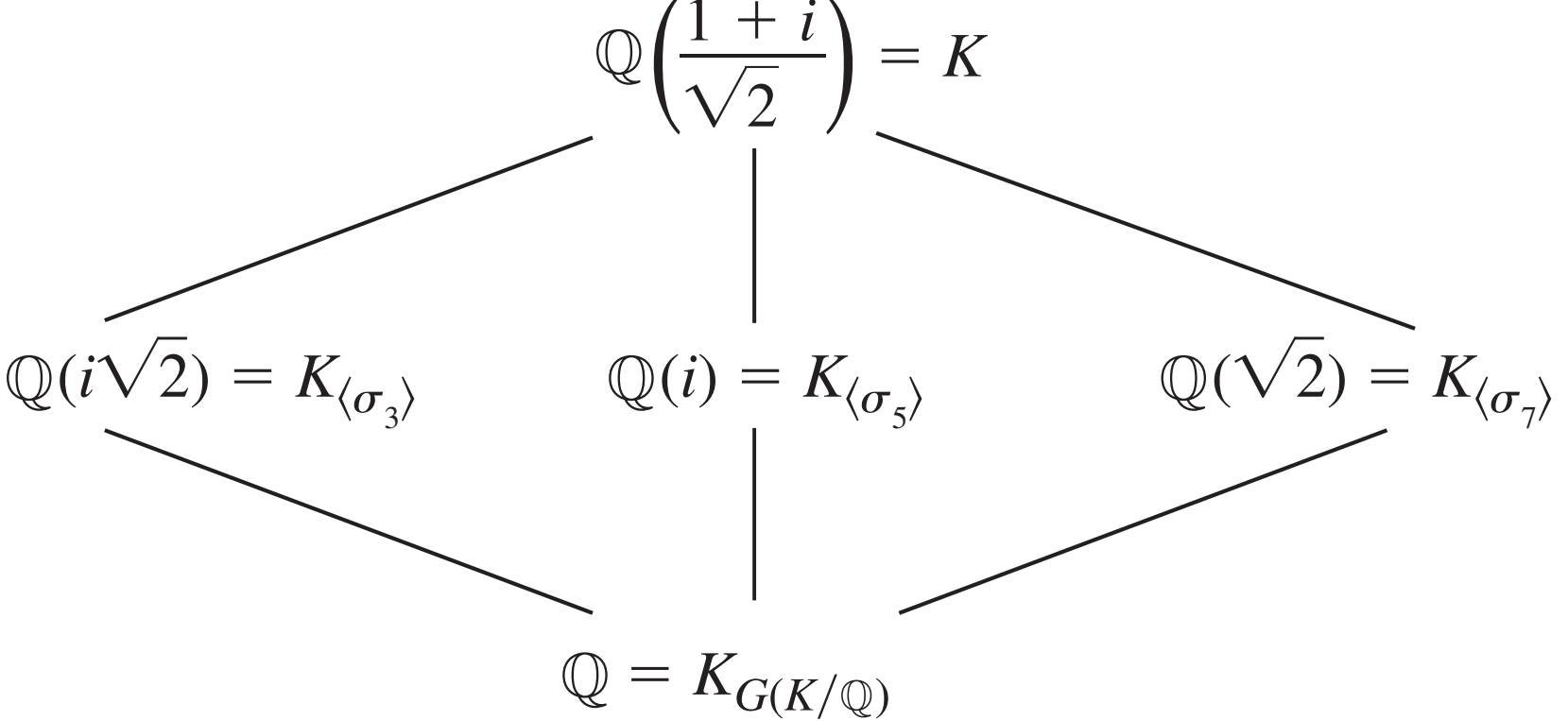

$\mathbb{Q}\left(\frac{1+i}{\sqrt{2}}\right) = K$
$\mathbb{Q}(i\sqrt{2}) = K_{\langle\sigma_3\rangle}$
$\mathbb{Q}(i) = K_{\langle\sigma_5\rangle}$
$\mathbb{Q}(\sqrt{2}) = K_{\langle\sigma_7\rangle}$
$\mathbb{Q} = K_{G(K/\mathbb{Q})}$

(b)